Advances in Insect Detection: A Comparative Survey of Models, Datasets, and Results

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I. Introduction

This report provides background for a future thesis on biodiversity monitoring, with a focus on reliable insect detection. It surveys recent progress across three connected themes: datasets (BIOSCAN-1M/5M, MassID45, and AMI), detection and segmentation models (from strong COCO-pretrained DETR-style baselines to insect-focused and edge-deployable pipelines), and the results reported under practical difficulties such as tiny targets, clutter, occlusion, and domain shift. A recurring motivation is BIOSCAN-style lab imagery, where each specimen image can be linked to DNA barcodes and hierarchical taxonomic labels, enabling multimodal identification and open-world evaluation.

II. Related Work

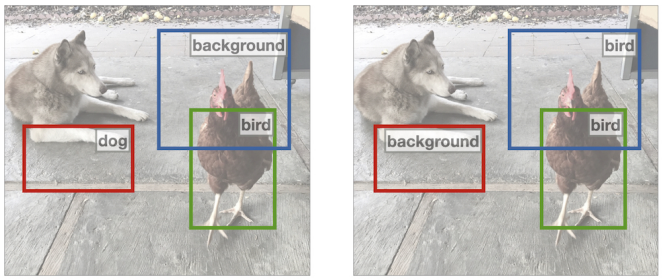
A. COCO Leaderboard Baseline: DETA-Swin-Large

A practical starting point for many detection problems is to fine-tune a strong COCO-pretrained model. In this report, DETA-Swin-Large (`jozhang97/deta-swin-large`) serves as a reference baseline because it is publicly available as a Hugging Face checkpoint [1] and is associated with strong COCO performance.

In *NMS Strikes Back* [2], DETA (*Detection Transformers with Assignment*) is presented. The main idea is to maintain the Deformable-DETR style architecture while reintroducing non-maximum suppression (NMS) during training and inference and substituting one-to-many IoU-based label assignment for one-to-one Hungarian matching. When compared to standard DETR-style training, the authors report strong COCO AP and faster convergence under matched training settings [2].

The official DETA repository also reports competitive results from smaller backbones. For transfer learning, the released `deta-swin-large` checkpoint is a high-capacity option (approximately 0.2B parameters) that can be fine-tuned as a strong localisation and feature-extraction starting point [1].

A practical example of this DETR family is RF-DETR, which packages a modern Transformer detector into an end-to-end pipeline for training, evaluation, and deployment. In line with recent comparisons, RF-DETR-style detectors typically combine strong self-supervised backbones (e.g., DINOv2) with deformable attention and end-to-end, NMS-free prediction, making them attractive when global context and occlusion handling matter. [3]



(a) One-to-one assignment (DETR-style)

(b) One-to-many assignment (DETA-style)

Fig. 1. Illustration of assignment strategies: one-to-one bipartite matching versus IoU-based one-to-many assignment used in DETA.

Relevance to insect detection. Insect imagery often contains small targets, cluttered backgrounds, and partial occlusions. Starting from a strong COCO-pretrained detector can provide robust low-level features and reliable localisation, which can then be adapted by fine-tuning on insect-specific data. In later sections, DETA-Swin-Large functions as a high-capacity reference point when comparing against insect-focused pipelines and deployment-oriented lightweight detectors.

Practical note (reproducibility). Hugging Face's Transformers documentation notes that DETA support has specific compatibility constraints; in practice, pinning a known working Transformers version helps avoid runtime issues [4].

B. RT-DETR: real-time end-to-end detection without NMS

RT-DETR is an end-to-end detector designed for real-time use without relying on NMS and its associated threshold tuning [5]. The paper motivates the approach by showing how NMS can add latency and make performance sensitive to confidence and IoU thresholds in common YOLO-style pipelines.

To make end-to-end detection practical at high speed, RT-DETR combines (i) an efficient hybrid encoder that separates intra-scale interaction from cross-scale fusion, and (ii) uncertainty-minimal query selection to initialise decoder queries with features that reflect both classification and localisation confidence [5]. A useful deployment feature is that speed can be adjusted by changing the number of decoder layers without retraining.

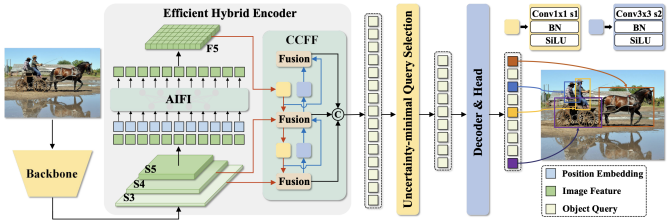


Fig. 2. RT-DETR overview (encoder–decoder design and query selection).

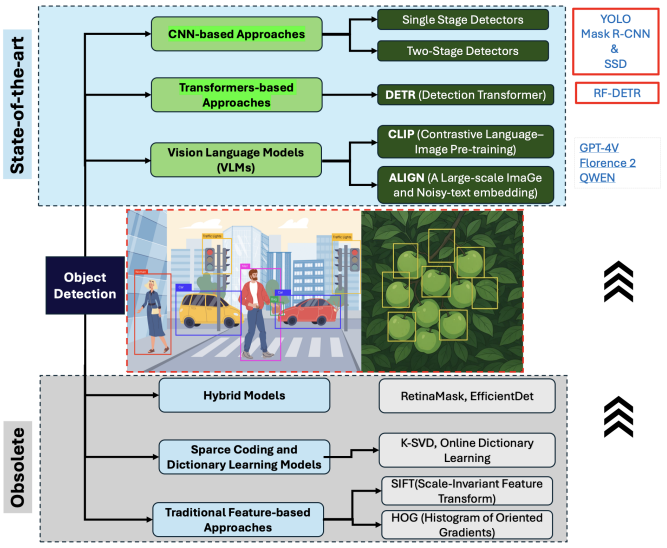


Fig. 3. Overview of the RF-DETR vs YOLOv12 evaluation setting in visually challenging orchard imagery.

Relevance to insect detection. RT-DETR is a useful reference when predictable latency matters and an end-to-end detector is preferred over NMS-based post-processing. Its encoder design and query selection are also relevant for small, cluttered targets where better initialisation can reduce missed detections under occlusion.

C. RF-DETR vs YOLOv12 in challenging field imagery

Sapkota et al. compare a transformer-based RF-DETR variant against YOLOv12 for green-fruit detection in orchard imagery where targets are often occluded, camouflaged, or difficult to label consistently [6]. Their model uses a DINOv2 backbone together with deformable attention, aiming to better capture global context when objects are only partially visible [6].

Reported results indicate that both detector families can be competitive, depending on the metric and setting. For single-class detection, RF-DETR achieves mAP@50 of 0.9464, while YOLOv12 variants remain strong on the stricter mAP@50:95 metric (e.g., 0.7620 for YOLOv12N) [6]. For multi-class detection, RF-DETR reaches the best reported mAP@50 of 0.8298, and YOLOv12L reports mAP@50:95 of 0.6622 [6]. The authors also note rapid convergence for RF-DETR in their setting, with training reaching a plateau in fewer than 10 epochs for the single-class case [6].

Relevance to insect detection. Although the dataset is agricultural rather than entomological, the visual conditions are similar to insect imagery in the wild (small targets, clutter, occlusion). The comparison helps motivate when transformer-based context modeling may improve robustness versus when a YOLO-style detector is preferable for edge constraints.

D. Comparison of DETA-Swin-Large, RT-DETR, and RF-DETR

This section compares the three object detection references used throughout the report: DETA-Swin-Large as a high-capacity COCO-pretrained transfer baseline, RT-DETR as an end-to-end detector designed for real-time use without NMS, and RF-DETR as a transformer detector evaluated under occlusion and label ambiguity.

Training and inference pipeline. DETA re-introduces non-maximum suppression (NMS) and uses one-to-many IoU-based assignment rather than one-to-one Hungarian matching, aiming for faster convergence and strong AP under comparable settings [2], [7]. In contrast, RT-DETR is explicitly motivated by removing NMS-related overhead and threshold sensitivity, and proposes architectural components (hybrid encoder and query selection) to make end-to-end detection practical in real-time settings [5]. RF-DETR is evaluated as a transformer-based alternative to YOLO-style pipelines for complex orchard imagery, emphasizing robustness under partial visibility and clutter [6].

Backbone and representational capacity. DETA-Swin-Large couples a high-capacity Swin-L backbone with a DETR-style detector and is released as a large pretrained checkpoint that is convenient for transfer learning [1], [7]. RT-DETR is presented with ResNet backbones (e.g., R50 and R101) and provides an explicit speed knob by varying decoder depth [5]. RF-DETR in the orchard study uses a DINOv2 backbone with deformable attention mechanisms to leverage global context when targets are partially occluded [6].

Speed–accuracy emphasis. RT-DETR reports high throughput with competitive COCO AP, highlighting the deployment-oriented trade-off between latency and accuracy (e.g., TensorRT FP16 benchmarking on an NVIDIA T4) [5]. DETA-Swin-Large prioritizes accuracy and transfer robustness, but its size suggests higher compute and memory requirements in embedded settings [1]. RF-DETR is not primarily a real-time paper; instead, it demonstrates that transformer-based context modeling can be advantageous in visually difficult conditions, while also showing that YOLO variants can be competitive depending on the metric (mAP@50 vs mAP@50:95) [6].

Implications for insect detection. Insect imagery shares properties with the challenging conditions emphasized by these works: small objects, clutter, and occlusion. A practical workflow is to treat DETA-Swin-Large as an upper-bound style baseline for transfer learning, use RT-DETR when predictable real-time behavior is needed without NMS tuning, and refer to RF-DETR-style comparisons when evaluating robustness under occlusion or ambiguous labeling. In later sections, this

TABLE I
QUALITATIVE COMPARISON OF THE THREE OBJECT DETECTION REFERENCES USED IN THIS REPORT.

Model	*Main characteristics (from the cited sources)*
DETA-Swin-Large	COCO-pretrained, high-capacity Swin-L checkpoint; DETR-style detector with one-to-many assignment and NMS re-introduced for strong AP and faster convergence; convenient transfer baseline but computationally heavy [1], [2], [7].
RT-DETR	End-to-end, real-time oriented detector designed to avoid NMS overhead; hybrid encoder and uncertainty-minimal query selection; speed can be tuned via decoder depth; strong COCO speed–accuracy trade-off reports [5].
RF-DETR	Transformer-based detector studied against YOLOv12 under occlusion and label ambiguity; uses a DINOv2 backbone and deformable attention to exploit context; highlights robustness trade-offs and metric sensitivity [6].

comparison motivates selecting either a high-capacity pre-trained detector for offline analysis, or a deployment-focused model for edge monitoring.

E. Insect detection and instance segmentation: flatbug

Svenning et al. introduce flatbug, a framework for detecting and segmenting terrestrial arthropods across diverse imaging systems and deployment contexts [8]. The central motivation is that arthropod appearance, scale, and density vary substantially across monitoring setups, and models trained for one environment often transfer poorly to another, leading to repeated manual labeling and retraining [8].

A key contribution is the flatbug dataset, which aggregates 23 subdatasets spanning both laboratory and field imaging regimes. In total, it contains 6,131 annotated images and 113,550 annotated arthropods, and is designed explicitly to support evaluation under domain shift across subdatasets [8]. The authors report strong validation performance for their best model (average F1 around 94%) and emphasise out-of-the-box generalisation using stratified cross-validation and leave-one-out evaluations [8].

flatbug builds on YOLOv8 instance segmentation and addresses a practical limitation of fixed-size inference when images are large and targets vary widely in scale [8]. During training, the method samples tile-like crops with controlled zoom levels and removes or inpaints partially visible instances to better match inference-time conditions. During inference, it applies a multi-scale pyramid tiling strategy so that both small and large arthropods can be detected reliably in high-resolution images.

Another defining design choice is a segmentation-first post-processing pipeline: flatbug performs deduplication using polygon IoU rather than bounding-box IoU, which is particularly important for closely packed or overlapping specimens [8]. To keep polygon-based overlap computation practical, the paper proposes a two-stage procedure that reduces the number of expensive polygon intersection operations while retaining accurate instance-level suppression.

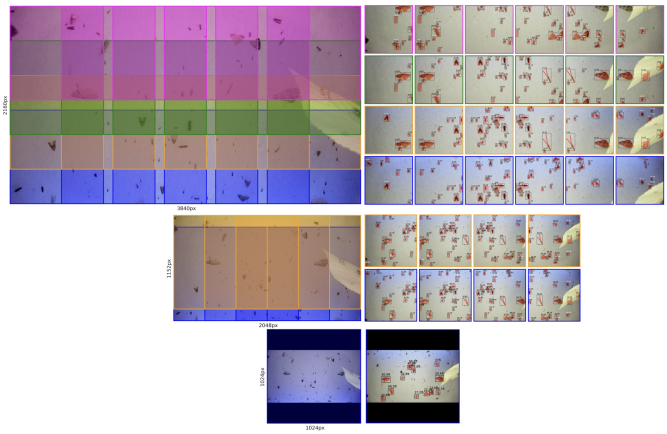


Fig. 4. flatbug multi-scale pyramid tiling strategy used to handle high-resolution images and small arthropod instances.

Relevance to this report. flatbug illustrates how insect-focused systems adapt general detectors through domain-aware engineering: dataset design, evaluation under domain shift, and scale-robust tiling. It also highlights that instance separation can be essential in practice, for example when downstream tasks require abundance estimates or size-related measurements.

F. CLIBD: bridging vision and genomics for biodiversity monitoring at scale

Gong et al. introduce CLIBD, a multimodal representation learning approach that aligns insect images, DNA barcodes, and text representations of taxonomic labels in a shared embedding space using CLIP-style contrastive learning [9]. The motivation is that image-only models often struggle with fine-grained species boundaries, while DNA provides a strong identity signal but is more expensive and less consistently available in deployment settings.

CLIBD uses three encoders (image, DNA, and text) initialised from pretrained models and fine-tunes them with a multimodal contrastive objective that aligns each modality pair within a batch [9]. At inference time, it supports nearest-neighbour style classification by embedding a query image and matching it against a database of embedded image and/or DNA records with known taxonomic labels. Because inference is retrieval-based, the reference database can be expanded over time by embedding new specimens, and the same model can support both seen and unseen species without task-specific fine-tuning [9].

Relevance to insect detection and BIOSCAN-style data. CLIBD directly targets the multimodal setting found in BIOSCAN-style pipelines, where lab images can be paired with barcode DNA and taxonomic labels. In an end-to-end

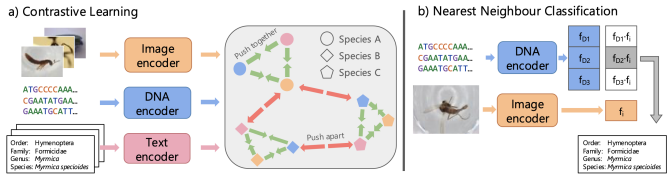


Fig. 5. CLIBD overview: multimodal alignment of insect images, DNA barcodes, and taxonomic text embeddings for retrieval-based identification.

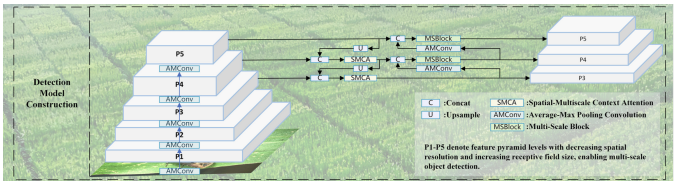


Fig. 6. AMS-YOLO architecture and modules (SMCA, MSBlock, AMConv) integrated into a YOLOv8n-style detector.

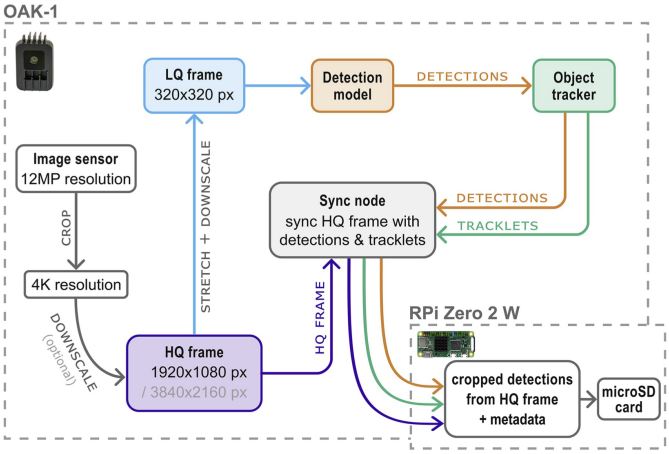


Fig. 7. Insect Detect system overview: edge camera trap components and on-device detection/tracking workflow for long-term field monitoring.

monitoring system, a detector can localize or crop insect instances, and CLIBD-style embeddings can then be used for downstream taxonomic identification, open-set retrieval, and cross-checking predictions with available DNA evidence. This makes CLIBD a close match to the data regime and objectives of the later thesis work.

G. Edge-friendly pest detection: AMS-YOLO deployed on Jetson Nano

Deng et al. propose AMS-YOLO, a lightweight enhancement of YOLOv8n for pest detection in complex field environments, targeting challenges such as inter-species visual similarity, background clutter, and large appearance changes across developmental stages [10]. The model introduces three modules: SMCA (spatial-multiscale context attention) to improve target recognition under complex backgrounds, MSBlock for multi-scale feature fusion, and AMConv for downsampling that preserves subtle discriminative details [10].

The paper evaluates on a 13-class maize pest dataset derived from IP102 plus additional field images, and reports improvements over the YOLOv8n baseline, reaching 94.2% mAP50 and 73.7% mAP50:95 while keeping the model compact (about 2.53M parameters and a 5.3MB weight file in their deployment report) [10].

A key feature for this report is the deployment study on an NVIDIA Jetson Nano. The authors describe a real-time monitoring pipeline with camera capture at 480×640 resolution and report stable-operation performance of 69.4 ms inference time, 81.7 ms total processing time, and 12.25 FPS, alongside reduced CPU utilization compared to YOLOv8n [10].

Relevance to insect detection on BIOSCAN-style data and future thesis work. Although AMS-YOLO is evaluated on agricultural pest imagery, the design goals align with insect monitoring workloads: small targets, cluttered backgrounds, and the need to run continuously on resource-constrained edge devices. For BIOSCAN-style pipelines, AMS-YOLO is a useful reference for the detection stage (localizing insects before downstream identification), and the Jetson Nano evaluation provides a concrete baseline for feasibility, latency, and model size when deploying detectors in the field.

H. Insect Detect: solar-powered edge camera trap with on-device detection and tracking

Sittinger et al. present Insect Detect, an open-source DIY camera trap for non-invasive monitoring of flower-visiting insects, designed explicitly for "AI at the edge" using low-cost off-the-shelf components and custom-trained deep learning models [11]. The system runs insect detection and tracking in real time on-device, and then performs a second-stage classification of cropped detections on a local computer.

The hardware design combines an OpenCV AI Kit (OAK-1) camera with an on-device inference chip, a Raspberry Pi Zero 2 W as host and storage controller, and a PiJuice Zero power-management board, powered by a small solar panel and batteries for season-long deployments [11]. A notable practical choice is the use of an artificial flower platform to standardize the background and insect posture, improving detection, tracking, and downstream classification.

The processing pipeline downscales frames to 320×320 for fast on-device detection, links detections across frames with a lightweight tracker, and synchronizes detections with high-resolution frames (1080p or 4K) to crop and store insect images with metadata on the device [11]. The authors report real-time operation around 12.5 FPS when synchronizing 1080p high-resolution frames, and they provide multiple YOLO-family detector variants deployable on the edge device.

Relevance to insect detection on edge devices and BIOSCAN-style workflows. Insect Detect provides a concrete end-to-end reference for field-ready insect monitoring: on-device detection and tracking reduce storage by saving only cropped detections and metadata, while enabling later higher-accuracy identification offline. This architecture maps well to BIOSCAN-style pipelines: an edge detector can generate instance crops in the field, and subsequent identification can be performed

Fig. 8. MassID45 example bulk image with dense, tiny arthropod instances and instance-level segmentation annotations.

using stronger classifiers or multimodal retrieval models (e.g., CLIBD) when barcode or label information is available.

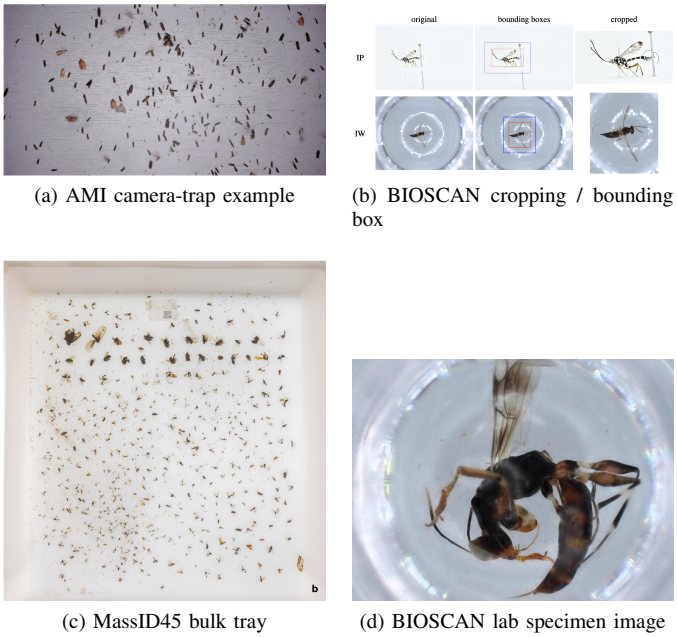
I. MassID45: multimodal bulk-sample dataset for tiny-instance segmentation and taxonomic labeling

Orsholm et al. present MassID45 (Mixed Arthropod Sample Segmentation and Identification), a dataset designed for unsorted bulk trap samples where individual insects appear as tiny, densely packed instances in high-resolution images [12]. The dataset is centred on 45 Malaise-trap samples collected in Sweden and Finland and provides paired information at two levels: bulk-sample imagery with DNA metabarcoding, and specimen-level images with individual DNA barcodes obtained after sorting [12].

At the bulk level, MassID45 includes 49 high-resolution bulk images (45 megapixels; 8192×5464) depicting unsorted specimens spread in a tray, with three metabarcoding replicates per sample [12]. Human annotators, supported by an AI-assisted tool, created instance segmentation masks and taxonomic labels for the bulk images, yielding 17,937 arthropod instance masks (class "bug") in COCO format. At the specimen level, the dataset provides 35,586 individual images and DNA barcode sequences, enabling construction of sample-specific taxonomies that constrain label choices during bulk annotation.

A key methodological contribution is the annotation workflow that combines (i) watershed-based initial masks, (ii) interactive mask refinement in the Toronto Annotation Suite (TORAS), and (iii) expert labeling guided by sample-specific taxonomies derived from specimen-level DNA barcodes [12]. The paper highlights practical challenges that are central to bulk insect imagery: most instances are extremely small (the majority fall into the "small" size regime under COCO-style evaluation), there is substantial debris, and many specimens overlap.

MassID45 is also positioned as a machine learning benchmark for tiny-instance segmentation. Because full bulk images do not fit typical GPU memory, the authors use tiling (512×512 with overlap) and merge predictions across tiles using slicing-aided hyper-inference (SAHI) with non-maximum merging [12]. They benchmark both zero-shot and supervised baselines for instance segmentation, reporting that supervised models fine-tuned on MassID45 substantially outperform zero-shot pipelines. For example, transformer-based Mask DINO



(a) AMI camera-trap example

(b) BIOSCAN cropping / bounding box

(c) MassID45 bulk tray

(d) BIOSCAN lab specimen image

Fig. 9. Representative dataset regimes discussed in this report: BIOSCAN lab imaging with detector-based cropping, MassID45 bulk trays with dense tiny instances, and AMI field camera-trap imagery with strong domain shift.

achieves mask $AP_{50:5:95}$ of 43.5 on the test set, compared to 27.1 for a Grounding DINO + SAM zero-shot pipeline [12].

Relevance to BIOSCAN-style data and future thesis work. MassID45 is closely aligned with the data regime of bulk insect monitoring and provides concrete guidance for scaling to BIOSCAN-style workflows: high-resolution bulk imaging, tiny-instance segmentation with tiling and robust duplicate-merging, and multimodal supervision where DNA-derived taxonomies help resolve fine-grained labels. In later sections, MassID45 serves as a reference for designing an end-to-end pipeline where detection/segmentation produces instance crops and abundance counts, while identification can be strengthened using barcode-linked databases.

III. Datasets

This report places particular emphasis on BIOSCAN-style lab imagery, where each specimen image can be linked to DNA barcodes and hierarchical taxonomic labels. The dataset section highlights how BIOSCAN (lab), MassID45 (bulk trays), and AMI (camera traps) cover complementary regimes that together resemble an end-to-end monitoring workflow.

A. BIOSCAN-1M Insect

Gharaee et al. introduce BIOSCAN-1M Insect as a large-scale, multimodal dataset pairing RGB microscope images with DNA barcodes and taxonomic annotations [13]. The dataset is dominated by long-tailed class imbalance and is commonly evaluated at coarser ranks (e.g., order) or on restricted subsets (e.g., Diptera families).

Data collection and modalities. Specimens were primarily collected from three countries (Costa Rica, Canada, and South

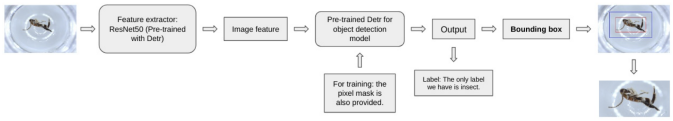


Fig. 10. BIOSCAN cropping pipeline: detector-predicted box is expanded to crop the specimen, then resized for efficient training.

Africa) using Malaise traps, and imaged with a Keyence VHX-7000 microscope [13]. The released metadata includes taxonomic labels, DNA barcode sequences, and indices linking each record to the corresponding image files and split membership [13].

Images and preprocessing. BIOSCAN-1M provides high-resolution RGB images (2880×2160) and multiple derived packages to support efficient training: original images (2.3 TB), cropped images (151 GB), resized originals with the smaller side set to 256 px (26 GB), and resized cropped images (7 GB) [13]. Cropping is performed by fine-tuning a DETR model on 2,000 annotated insect images, then selecting the highest-confidence predicted box and extending it before cropping; the cropped image is finally resized to 256 px on its smaller side for learning efficiency [13].

B. BIOSCAN-5M

BIOSCAN-5M expands BIOSCAN-1M into a broader, cleaned, and more diverse multimodal resource, containing 5,150,850 arthropod specimens (about 98% insects) and substantially more fine-grained labels, including around 1.2 million records labelled to genus or species taxonomic ranks [14]. Relative to BIOSCAN-1M, it increases geographic coverage (3 countries to 47) and taxonomic breadth (from a single class and 16 orders to 10 classes and 55 orders), and it adds geographic and specimen-size metadata that can be used as auxiliary signals [14].

Images and cropping tool. BIOSCAN-5M provides specimen images at 1024×768 pixels, captured using a Keyence VHX-7000 microscope [14]. Following BIOSCAN-1M, images are cropped and resized to approximately 341×256 for model training, using a DETR-based cropping tool (with additional fine-tuning to address previously poorly cropped cases) and releasing the predicted bounding boxes as part of the dataset [14].

Metadata fields and available packages. In addition to the taxonomic hierarchy and DNA fields (e.g., dna_barcode and dna_bin), BIOSCAN-5M provides country/province and coordinates, and size-related fields such as area_fraction and scale_factor [14]. The dataset is distributed with multiple image packages (original_full, cropped, original_256, cropped_256) and predefined partitions for both closed-world and open-world evaluation settings [14].

C. MassID45 bulk-sample imagery with DNA at trap and individual level

Orsholm et al. present MassID45 (Mixed Arthropod Sample Segmentation and Identification), a multi-modal dataset

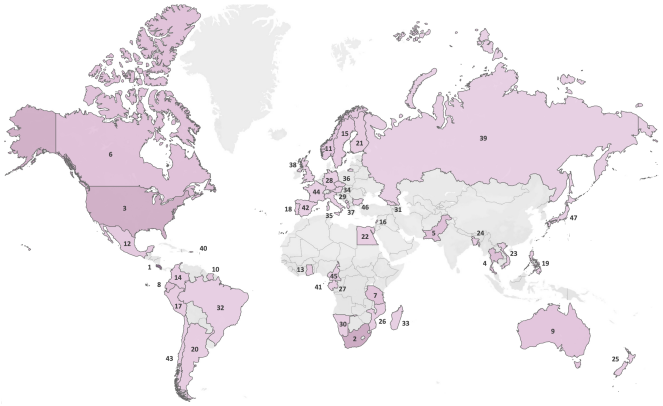


Fig. 11. BIOSCAN-5M overview: expanded scale and diversity, with multimodal metadata for biodiversity monitoring.

designed around unsorted bulk trap samples, where thousands of tiny arthropods appear densely packed in high-resolution images [12]. The dataset provides paired information at two complementary levels: bulk-sample images with DNA metabarcoding for taxonomic composition, and specimen-level images with individual DNA barcodes obtained after sorting [12].

Bulk images and annotations. MassID45 includes 49 edited bulk images (45 megapixels; 8192×5464) from 45 Malaise-trap samples (4 samples have two images), captured in a controlled light-cube setup with a top-down camera view [12]. The authors provide instance segmentation masks for 17,937 arthropods in COCO format and taxonomic labels with high- and low-confidence annotation mechanisms to express uncertainty across ranks [12].

DNA and specimen-level records. Each bulk sample has three metabarcoding replicates (short COI fragment), and the dataset also provides 35,586 individual specimen images and corresponding DNA barcodes (full COI region) from the same samples [12]. This design supports experiments that combine image-based abundance estimation with DNA-informed taxonomic resolution.

ML implications. Because full images exceed typical GPU memory, the paper recommends tiling-based training and inference (e.g., 512×512 with overlap) and merging predictions across tiles using SAHI and non-maximum merging [12]. This makes MassID45 a practical benchmark for tiny-instance segmentation and bulk-sample pipelines.

D. AMI: insect identification in the wild with strong domain shift

Jain et al. introduce the AMI (Automated Monitoring of Insects) dataset as a benchmark for fine-grained insect recognition under realistic field conditions, with an explicit emphasis on distribution shift between human-captured images and automated insect camera traps [15]. The motivation is that in-the-wild insect monitoring combines long-tailed species distributions, extremely similar classes, and significant shifts in imaging hardware, lighting, and insect pose, while expert

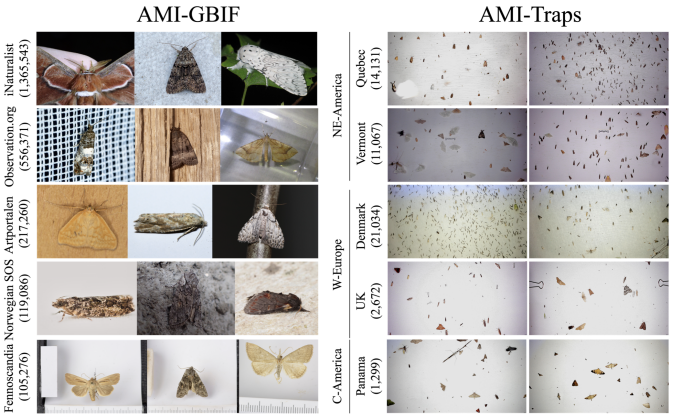


Fig. 12. AMI domain shift setting: curated images versus camera-trap conditions (partial insects, overlap, blur, and lighting variation).

annotation for trap data is expensive and geographically localized [15].

AMI is composed of two complementary parts. AMI-GBIF contains approximately 2.5M images curated from citizen science platforms and museum collections, covering 5,364 moth species across 1,734 genera and 77 families (plus an additional binary moth vs non-moth dataset) [15]. AMI-Traps is an expert-annotated dataset of 2,893 camera trap images collected across five countries (Denmark, UK, Quebec/Canada, Vermont/USA, and Panama), with 52,948 labeled insects and fine-grained moth labels where possible [15]. The paper highlights practical field challenges such as partial insects, overlap, blur, variable lighting, and low resolution.

ML implications. AMI is primarily a fine-grained classification and domain generalization benchmark rather than an object detection benchmark; however, the authors describe a realistic pipeline where detection is used to produce insect crops for identification. To mitigate the domain gap between high-resolution training data and low-resolution trap crops, they introduce a simple mixed-resolution augmentation (MixRes) that improves out-of-distribution performance on AMI-Traps while maintaining strong in-distribution accuracy [15].

Relevance to this report. AMI complements BIOSCAN-style lab datasets by emphasizing field deployment conditions and domain shift. It provides a useful reference when connecting an edge or trap-based detection stage (e.g., Insect Detect / AMS-YOLO) to downstream identification models that must remain robust across hardware and geography.

E. Implications for this preliminary study

BIOSCAN-1M motivates robust detection and cropping for small, variable specimens under heavy class imbalance, while BIOSCAN-5M provides the scale, diversity, and metadata (DNA, taxonomy text, geography, size) needed to study multimodal identification and open-world evaluation. MassID45 complements these resources by focusing on unsorted bulk samples with dense, tiny instances, providing instance masks and DNA at both trap and specimen level; it is therefore a useful reference for designing tiling-based detection/segmentation

(a) Jetson Nano (b) Raspberry Pi Zero 2 W

(c) Luxonis OAK-1 (d) Coral USB Accelerator

Fig. 13. Representative edge hardware used in field monitoring pipelines: (a) an NVIDIA Jetson platform for GPU-accelerated inference, (b) a Raspberry Pi host/controller, (c) an OAK smart camera with on-device VPU inference, and (d) a Coral USB Accelerator for low-power Edge TPU inference. Photo sources: [16]–[19].

and downstream identification workflows. Finally, AMI provides a field-oriented benchmark for fine-grained recognition under strong domain shift from human-captured imagery to automated camera traps, helping motivate robustness and domain adaptation considerations for deployment. In the remainder of this report, these datasets define the target data regime for the later thesis work.

IV. Edge Computing Devices for Insect Detection in the Field

Edge computing allows insect detection to run close to the data source (e.g., camera traps, light traps, or monitoring stations), reducing latency, bandwidth requirements, and reliance on continuous connectivity. This is particularly valuable in remote or power-constrained settings, where devices must operate autonomously for long periods.

A. Design constraints for field deployment

Edge devices used for insect detection are constrained by several factors: limited power budgets (often battery or solar powered), restricted memory and compute, environmental robustness, and the need for reliable long-term operation. These constraints strongly influence model choice, favoring lightweight detectors, reduced input resolutions, and architectures optimized for efficient inference.

B. NVIDIA Jetson platforms

NVIDIA Jetson devices (e.g., Jetson Nano, Xavier NX, Orin Nano/NX) provide GPU-accelerated inference and strong support for modern deep learning frameworks. Jetson Nano has been widely used as a baseline edge platform in insect and pest detection studies, including the deployment of AMS-YOLO, where real-time performance around 12 FPS was achieved with a compact YOLOv8n-based model. More powerful Jetson variants (Xavier NX, Orin) support larger backbones and higher throughput, enabling deployment of more accurate but computationally heavier detectors. The trade-off is increased power consumption, which may limit suitability for long-term solar-powered traps.

C. Raspberry Pi-based systems

Raspberry Pi devices are commonly used as low-cost edge controllers in insect monitoring systems. While CPU-only inference limits achievable frame rates, Raspberry Pi boards are effective when paired with lightweight models or external accelerators. They are frequently used for system orchestration tasks such as camera control, data storage, scheduling, and communication, rather than as the primary inference engine. Their low idle power consumption and broad peripheral support make them attractive for field deployments.

D. Vision accelerators and smart cameras

Dedicated vision accelerators offer a favorable balance between performance and energy efficiency. OpenCV AI Kit (OAK) cameras integrate an Intel Movidius VPU with on-device neural inference and have been successfully deployed in solar-powered insect camera traps. In such systems, lightweight YOLO-family detectors run directly on the camera at around 10–15 FPS, while only cropped detections and metadata are stored or transmitted. This architecture substantially reduces power usage and data volume.

E. Edge TPUs and low-power accelerators

Google Coral Edge TPU devices provide efficient inference for quantized models and are suitable for continuous monitoring scenarios. When combined with lightweight detectors (e.g., MobileNet- or YOLO-tiny-style architectures), Edge TPUs enable real-time detection at very low power consumption. Model compatibility and quantization requirements impose constraints, but the resulting energy efficiency makes these devices well suited for large-scale distributed trap networks.

F. Model–device matching

The choice of edge device is closely tied to the detection model. High-capacity models (e.g., transformer-based detectors or instance segmentation pipelines such as flatbug) are better suited to GPU-enabled platforms or offline processing, whereas lightweight YOLO-based detectors (e.g., AMS-YOLO, Insect Detect variants) are well matched to VPUs and TPUs. In practical deployments, a common strategy is to perform detection on the edge using an efficient model, and defer fine-grained identification or multimodal analysis (e.g., CLIBD-style retrieval) to offline or centralized systems.

TABLE II
PRACTICAL DEVICE–MODEL MATCHING FOR FIELD INSECT MONITORING (QUALITATIVE GUIDANCE).

Device class	*Typical fit for models in this report*
Jetson (GPU edge)	Supports heavier detectors and higher throughput. Suitable for medium-size YOLO variants and for real-time deployment-focused models when power budget allows. Can also run larger backbones at reduced frame rate when accuracy is prioritized.
Smart camera / VPU (e.g., OAK)	Best for efficient detectors where the camera performs inference and the host handles orchestration. Good match for lightweight YOLO-style detectors and tracking + crop saving pipelines.
Edge TPU / low-power accelerator	Strong energy efficiency for quantized models. Best for continuous monitoring with compact architectures; often paired with a small host (e.g., Raspberry Pi) for storage and communication.
Raspberry Pi (CPU host)	Best as a controller for capture, scheduling, storage, and networking; can run very small models, but commonly paired with an accelerator for practical frame rates.

G. Implications for this study

For BIOSCAN-style workflows and future thesis work, edge devices primarily serve as the first-stage detection component, generating reliable insect crops under field conditions. The reviewed platforms demonstrate that meaningful trade-offs between accuracy, throughput, and energy consumption are achievable, and that careful co-design of models and hardware is essential for scalable insect monitoring in real-world environments.

V. CONCLUSION

This report surveyed recent work that informs practical insect detection systems: general-purpose detection baselines, insect-focused detection and segmentation pipelines, multimodal identification methods, large-scale datasets, and field-ready edge deployments.

Across model families, transformer-based detectors such as DETA and RT-DETR provide strong accuracy and well-motivated training and inference design choices, while lightweight YOLO-style models remain attractive when compute and latency dominate deployment constraints. The

insect-focused systems reviewed here (flatbug and Insect Detect) show that domain-aware engineering—tiling strategies, instance-centric post-processing, tracking, and annotation workflows—often matters as much as the backbone architecture.

On the data side, BIOSCAN-1M and BIOSCAN-5M establish a regime where specimen images can be linked to DNA and hierarchical taxonomy, enabling multimodal learning and open-world evaluation at scale. MassID45 complements this by focusing on unsorted bulk samples and tiny-instance segmentation, highlighting the need for tiling and careful merging of overlapping predictions. AMI emphasises that field deployment introduces strong domain shift from curated imagery to camera-trap conditions, motivating robustness and adaptation strategies.

Finally, the edge-device perspective connects these components into an end-to-end monitoring pipeline: efficient detectors can run on-device to produce reliable instance crops and metadata, while heavier identification models and multimodal retrieval (e.g., CLIBD) can be applied offline or centrally when additional compute or DNA evidence is available. Together, these findings provide a grounded foundation for subsequent thesis work, with clear criteria for selecting datasets, model families, and deployment hardware based on the target monitoring scenario.

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